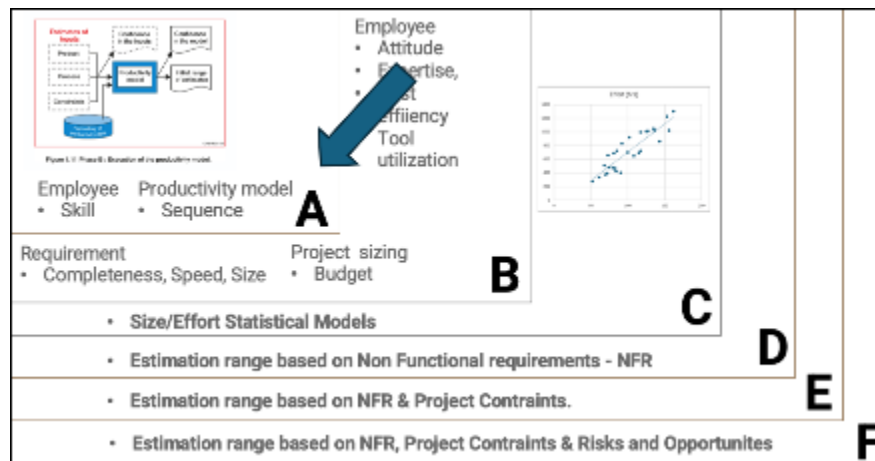


Software Projects Estimation (Educational Games)

Module: Estimation Planning Overview

(Version: June 22, 2026)

Audience: Students



Editor: Alain Abran

Key contributors: Ling Guo, Sion Israel Sion, JM Desharnais, Monica

Villavicencio

Sponsors:



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1. Introduction/Background

This educational game teaches players how to select estimation team members based on the skills required to perform the statistical steps involved in building a linear regression model of the size–effort relationship.

The game requires the player to choose an employee whose skills match the dataset of software projects being analyzed. The player must also analyze possible combinations of analysis steps and select among ten alternative sequences for constructing a linear regression model using the dataset.

Through this process, the game illustrates that the sequence of steps for building a linear regression model may vary depending on the characteristics of the dataset, such as when software size information is missing or when statistical outliers are present.

2. How to Use this Module

This module is designed to be used within a training course in software project management. It is intended to be used as a *complementary* component to a course, not as a standalone instructional tool.

Since the estimation approach used in this educational game is based on the software size in Function Points, software functional sizing must be introduced to students either before, or in parallel with the students' exposure to the game (see the References, or some other method).

Before students are given the assignment to play a SimSE game, it is imperative that they watch the SimSE Gameplay video tutorial. It is strongly recommended that they also watch the Explanatory Tool and Game Branching video tutorials as well. All video tutorials are available at <http://www.ics.uci.edu/~emilyo/SimSE/downloads.html#Videos>.

Our experience with SimSE has shown again and again how crucial the instructions a student receives in learning to play the game are to their success in learning from it.

These video tutorials have been designed to specifically highlight and address aspects of SimSE that are critical for students to understand for a maximally effective educational experience. Therefore, we suggest that you not only assign the students to watch these videos on their own time, but, if time and resources allow, show them in class as well, emphasize how important they are to watch, and also point them to the SimSE player's manual, available at:

<http://www.ics.uci.edu/~emilyo/SimSE/downloads.html#Docs>.

If time and resources further warrant, students should be required to attend a teaching-assisted-lead training session, in which they are shown the videos, given a printed player's manual and, and then allowed to try playing the game for a while with the Teaching Assistant (who should have already studied the manual and played SimSE themselves) available to answer any questions they may have).

Students should be given the questions to answer this module (see Section 8) at the time they are asked to play the game. Having the questions to refer to while they play helps point them to some of the more subtle lessons encoded in the game, as well as provides you, as an instructor, with a way to assess whether or not they have completed the assignment and learned the concepts.

One could use this module as a mandatory part of a course, or else make it an extra-credit assignment. Given that the exercise is actually quite involved, especially when multiple models are assigned, somewhere in the order of 5-10% extra credit is recommended. Certain prerequisites for this particular module apply (see Section 4).

3. Learning Objectives

The learning objectives are from a software management and statistical analysis perspective. The following three game-specific learning objectives are implemented in this game:

- *The students should know that some steps in building a linear regression model may be repeated multiple times.* When datasets with different characteristics are used for a statistical regression analysis, there may be a different sequence of steps: for example, the identification and removal of statistical outliers before a regression analysis may require several iterations.
- *The students must take into account a) the type of data sets used; b) the necessary employee skills for the datasets selected.* For example:
 - o For a data set of in-house software projects for which the software size in Function Points is not available, the measurement of their software size in Function Points is required before proceeding to a linear regression analysis.
 - o Such a measurement step is not necessary for the ISBSG dataset which already has for all projects their size in Function Points.

4. Prerequisites

For this game, players should have a generic knowledge of linear regression techniques and Function Points (Function Points Analysis – ISO 20926, COSMIC – ISO 19761, NESMA, etc.). The main idea of this game is to take management decisions on resources selection and corresponding statistical analysis activities for a specific data set of software projects.

5. Time Commitment

The average time to play a single game is 10-15 minutes, but, of course, it is likely to take several iterations playing the game for the student to learn the concepts and be able to answer the questions. Players should be given at least one week of out-of-class time to explore the game and answer the questions (see Section 8).

6. Instructions

First, watch the SimSE video tutorials at:

<http://www.ics.uci.edu/~emilyo/SimSE/downloads.html#Videos>.

Then download the SimSE player's manual at:

<http://www.ics.uci.edu/~emilyo/SimSE/downloads.html#Docs>.

Be sure to watch the video and read the manual carefully, as they will highlight several important things that will significantly help you in successfully playing SimSE and correctly answering the questions.

Next, download the game at:

<http://www.ics.uci.edu/~emilyo/SimSE/downloads.html#Games>. The download consists of a “readme” text file and an executable game, which you can run by simply double-clicking on it. If you do not have the current version of Java installed on your machine, you will have the opportunity to install it when you try to run a game.

7. Questions

We recommend answering the following two questions with this module.

1. When doing a statistical regression analysis of software projects using software size as the independent variable, what is a key difference between in-house projects and the projects available in the ISBSG dataset – (ISBSG: International Software Benchmarking Standards Group – isbsg.org)?
2. Why does the sequence of steps for a statistical linear regression analysis vary?

8. Other notes

There are several other potentially effective uses for SimSE:

- Have students play in teams, especially teams that have also done, or are doing, a class project together. This can add both a collaborative aspect to learning and, if set up to be a competition between teams, can add a competitive aspect.
- Make the assignment mandatory, rather than optional or extra-credit, to increase motivation.
- Have students play in a lab setting, both to add a competitive aspect and to allow them to collaborate.
 - o Keep in mind, however, that a lab setting generally does not provide enough time to play a game enough to be able to answer all the questions.
 - o An appropriate approach might be to allow students to play the game first in a lab session (this would also allow them to ask any questions that may arise) and then let them complete the rest of their playing and question-answering out of class.

9. Feedback?

If you have any comments, suggestions, feedback, or experience regarding this course module, please send an email to Alain Abran: alain.abran@etsmtl.ca

References

1. Navarro, E.O. “*The Fundamental Rules*” of Software Engineering. 2008 [Available from: http://www.ics.uci.edu/~emilyo/SimSE/se_rules.html].

2. [Introduction to the COSMIC Software Sizing Methodology cosmic-sizing.org](https://cosmic-sizing.org)
3. COSMIC. (2020). Early Software Sizing with COSMIC: Practitioners Guide.
4. COSMIC. (2020). Early-Sizing-Experts-Guide-May-2020-1. <https://doi.org/10.13140/RG.2.1.4195.0567>
5. Alain. Abran, (2015). Software Project Estimation: The fundamentals for providing high quality information to decision makers. John Wiley & Sons.